

DONGWON JEONG

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RESEARCH INTERESTS

Wearable Robotics, Biomechanics, Human-Robot Interaction (HRI), Reinforcement Learning, Rehabilitation

EDUCATION

- **Yonsei University** Sep 2026 - Aug 2028(expected)
Seoul, Republic of Korea
M.S. Medicine
◦ GPA: TBD
- **Yonsei University** Mar 2019 - Aug 2026
Seoul, Republic of Korea
B.S. Bio-Convergence & B.S. Computer Science and Engineering
◦ GPA: 3.64/4.0
- **Georgia Institute of Technology** Aug 2025 - Dec 2025
Atlanta, United States
Exchange Student / College of Computing, Computer Science
◦ GPA: 4.0/4.0
- **Ecole International Bilingue - Victor Hugo School** Sep 2015 - Jun 2018
Paris, France
International Baccalaureate Student
◦ Salutatorian, Student Vice President(2017-2018)
◦ SAT: 1560/1600, IB: 42/45

RESEARCH EXPERIENCE

- **BioMechanics and Robotic Rehabilitation Laboratory (BMRR Lab), Yonsei Univ.**  Feb 2025 - Present
Research Assistant Advisor: M.D. Dongwook Rha
◦ Investigated the aftereffect of gait CIMT(Constraint Induced Movement Therapy) using a robotic hip exoskeleton
◦ Audited 'MED9079: Motion Analysis and Human Gait' course of Yonsei University Graduate College of Medicine
- **Reinforcement Learning and Robot Learning Lab (RLLAB), Yonsei Univ.**  Mar 2026 - Jun 2026
Undergraduate Capstone Design Researcher Advisor: Ph.D. Youngwoon Lee
◦ Capstone Design Topic: Large-Scale Parallel Reinforcement Learning for Humanoid Dexterous Manipulation
◦ Investigated improvements over PPO (e.g., EPO, SAPG) for large-batch parallel policy optimization in humanoid control tasks, extending prior work validated on fixed-base manipulators
◦ Industry-Academia Collaboration project with Amazon Web Services (AWS)
- **Exoskeleton and Prosthetic Intelligent Controls Lab (EPIC Lab), Georgia Tech.**  Aug 2025 - Dec 2025
Undergraduate Research Assistant Advisor: Ph.D. Aaron Young
◦ Collected and processed post-stroke multimodal gait data (IMUs, pressure insoles, motion-capture)
◦ Developed a scripted pipeline (segmentation, normalization, low-pass filtering, exception handling) to process post-stroke gait data and conduct cross-correlation analysis of gait metrics
◦ Implemented an effective real-time visual biofeedback system targeted at achieving positive hip work symmetry and designed experimental protocols for upcoming clinical validation studies
- **High Performance Computing Platforms Lab (HPCP Lab), Yonsei Univ.**  Mar 2025 - Jun 2025
Undergraduate Capstone Design Researcher Advisor: Ph.D. Youngsok Kim
◦ Capstone Design Topic: Accelerating RDBMS with CPU Index Filtering for Efficient GPU Query Execution
◦ Industry-Academia Collaboration project with Samsung Electronics
- **Computational Neuroscience Lab (cNeuro Lab), Yonsei Univ.**  Dec 2022 - Jun 2023
Undergraduate Research Assistant Advisor: Ph.D. Sehun Chun
◦ Studied Spiking Neural Network (SNN) and neural coding fundamentals
◦ Implemented MATLAB codes to simulate action potential propagation of different neuron models
◦ Participated in the 2023 Korean Society for Human Brain Mapping (KHBM) Winter School
◦ Participated in the 2023 Korean Society for Computational Neuroscience (KSCN) Winter School

WORK EXPERIENCE

- **AidALL  | Korea Institute of Science and Technology (KIST)** Jan 2026 - Feb 2026
Robotics & AI Engineer Intern Seoul, Republic of Korea
 - Designed and implemented an instrumented walker platform integrating IMUs, force sensors, and motion capture data as preliminary research toward the development of smart assistive mobility systems.
 - Deployed and debugged an NVIDIA Isaac ROS visual SLAM and nvblox mapping pipeline using Intel RealSense depth cameras, resolving configuration, synchronization, and runtime issues to enable stable 3D reconstruction.

AWARDS & SCHOLARSHIPS

- **Yonsei Underwood International College (UIC) Scholarship** Aug 2025
Yonsei University, Underwood International College
 - \$1.6k Granted for participation in outbound exchange student program
- **Yonsei University Computer Capstone Design Excellence Award** Jun 2025
Yonsei University, College of Computing | National Center of Excellence in Software
 - Selected as one of 4 teams out of 15 that exhibited at the 2025 Spring CS Senior Research Exhibition
- **US-Korea High-Tech Industry Exchange Scholarship** Jun 2025
Republic of Korea Ministry of Trade, Industry & Energy | Korea Institute for Advancement of Technology (KIAT)
 - \$9k Granted for acknowledgment of highly exceptional academic achievement and potential
- **2024 Korea Army Startup Competition Creativity Award** Oct 2024
Republic of Korea Army, Korea Army Personnel Command 
 - Selected as 25/315 qualified finalist of 2024 Korea Army Startup Competition
- **Yonsei Eagle Tutoring Scholarship** Jun 2023
Yonsei University, Innovation Center for Teaching and Learning
 - \$450 Granted for successfully mentoring 4 undergraduate tutees throughout the semester
- **Yonsei Python Data Analysis Competition 2nd Prize** Feb 2023
Yonsei University, Office of Library and Cultural Services
 - Conferred for being 2nd place in a Python Data Analysis Competition
- **2018 Mathematics Olympiad for Koreans in France 1st Prize** May 2018
Association des Scientifiques Coréens en France (ASCOF)
 - Conferred for being 1st place in a national-scale Mathematics Olympiad
- **2016 Mathematics Olympiad for Koreans in France 2nd Prize** May 2016
Association des Scientifiques Coréens en France (ASCOF)
 - Conferred for being 2nd place in a national-scale Mathematics Olympiad

MILITARY EXPERIENCE

- **Korea Defense Intelligence Command | Defense Intelligence Agency** Aug 2023 - Feb 2025
Military Interpreter | Sergeant | Squad Leader Anyang, Republic of Korea
 - Translated classified & unclassified military intelligence documents, reports, and articles
 - Awarded 4 medals each conferred by *Korea Defense Intelligence Command Commander (Brigadier General)*, *Korea Army Personnel Command Commander (Major General)*, *Korea Army Training Center Commander (Major General)*, *Korea Army Training Center Regiment Commander (Colonel)*
 - Cleared for 2nd highest Classified Information Clearance
 - Co-published the Korean version of 'The Military Balance 2024,' an open-source assessment of the military forces, personnel numbers, equipment inventories, and defense economics of over 170 countries
 - Co-published 'Foreign Intelligence Report on North Korea 2023,' 'Foreign Intelligence Report on North Korea 2024,' and 'Foreign Intelligence Report on North Korea 2025' which are comprehensive annual reports on North Korea
- **United Nations Industrial Development Organization - ITPO Korea ** Mar 2024 - Jun 2024
Technical Interpreter | Corporal Seoul, Republic of Korea
 - Dispatched to *International Security Exhibition & Conference (SECON)* and *e-Government Information Security Solution Fair(eGISEC) 2024* to assist an Ecuadorian Navy Security Department personnel consult and negotiate business which aimed at controlling the 2024 Ecuadorian internal armed conflict and security of an Ecuadorian naval base

TEACHING EXPERIENCE

- **ISE Biochemistry(1) / Yonsei University** Sep 2022 - Dec 2022
Tutor Seoul, Republic of Korea
 - Held the role of tutor to five tutees
- **SATUS(International Baccalaureate Education Institute)** Jun 2020 - Jan 2021
IB Instructor Seoul, Republic of Korea
 - Gave lectures on IB Biology, Chemistry, and Mathematics (Standard & Higher Level)

VOLUNTEER SERVICE

- | | |
|---|--|
| • Undergraduate Volunteer Service
130.5 Hours | Mar 2019 - Present
Seoul, Republic of Korea |
| • High School Volunteer Service
241 Hours | Sep 2015 - Jun 2018
Paris, France |

EXTRACURRICULAR ACTIVITIES

- | | |
|--|---|
| • Seojo FC
President
◦ Organized and participated in weekly Saturday futsal game sessions | Aug 2023 - Present
Seoul, Republic of Korea |
| • Yonsei Computer Club (YCC)
Member
◦ Participated in Computer Vision study group (2025 spring)
◦ Participated in Swift/iOS study group (2023 spring)
◦ Lead an advanced C++ study group (2023 spring)
◦ Participated in the AI bootcamp hosted by YCC (2022 winter)
◦ Lead a basic C++ study group (2022 winter) | Sep 2022 - Aug 2025
Seoul, Republic of Korea |
| • Google Developer Student Clubs(GDSC) Yonsei University Seoul Campus
Developer
◦ Joined after a competitive selection process as one of 16 developers with a focus on AI and iOS
◦ Participated in a computer network lab study using Wireshark
◦ Implemented a TCP-based chat application with custom application protocol
◦ Worked as event staff at the 2023 Google I/O Extended Seoul session | Mar 2023 - Aug 2023
Seoul, Republic of Korea |
| • Yonsei University CSI4108 (Artificial Intelligence) Study Group
Group Leader
◦ Lead the CSI4108 (Artificial Intelligence) course study group | Mar 2023 - Jun 2023
Seoul, Republic of Korea |
| • Yonsei University Embedded System Club (BitCrab)
Member
◦ One of the founding members of the first embedded system club of Yonsei University | Mar 2023 - Aug 2023
Seoul, Republic of Korea |
| • Mensa France Developer Algorithm Study Group
Member
◦ Solved algorithm problems in leetcode.com, acmicpc.net, and swexpertacademy.com
◦ Interacted online with other members to share solutions | Aug 2022 - Present
Paris, France |

PUBLICATIONS & PRESENTATIONS

S=IN SUBMISSION, C=CONFERENCE, O=ORAL PRESENTATION, B=BOOK

- [C.3 & S.1] D. Jeong, J. Hong, J. Yoo, J. Cha, T. Kim, Y. Kim, D.-W. Rha. (2026). "Assessing Gait and Gross Motor Function in Cerebral Palsy from RGB Video Using AI Pose Estimation" 2026 Korean Academy of Rehabilitation Medicine (KARM) International Fall Conference.
- [C.2 & O.2] D. Jeong, S. Oh, S. Jeong, W. Kim. (2026). "Multimodal Analysis of Sequencing-Dependent Coordination in Instrumented Walker-Assisted Gait" 7th International Conference on NeuroRehabilitation (ICNR 2026).
- [C.1 & O.1] D. Jeong, D. Park, A. Young. (2026). "Biomechanical Correlates of Positive Hip Joint Work and Symmetry in Post-Stroke Gait". 10th World Congress of Biomechanics (WCB 2026).
- [B.5] D. Jeong, et al. (2026). "Foreign Intelligence Report on North Korea 2025". Korea Defense Intelligence Command
- [B.4] D. Jeong, et al. (2025). Korean Version: "The Military Balance 2024". Korea Defense Intelligence Command
- [B.3] D. Jeong, et al. (2025). "Foreign Intelligence Report on North Korea 2024". Korea Defense Intelligence Command
- [B.2] H. Jeong, D. Jeong. (2024). "The Dream of Baby Arrowroot". Samil. ISBN 9791189942311
- [B.1] D. Jeong, et al. (2024). "Foreign Intelligence Report on North Korea 2023". Korea Defense Intelligence Command

PROJECTS

- Markerless Gait Analysis for Cerebral Palsy Using AI Pose Estimation**
Tools: [Python, RTMPose, VICON]
 - Developed a markerless gait analysis pipeline using RTMPose to estimate lower-limb kinematics from ordinary RGB clinical videos and extract hip-knee cyclogram and hip phase-portrait gait biomarkers.
 - Validated video-derived biomarkers against 3D motion capture and demonstrated significant correlations with Gross Motor Function Measure (GMFM-66) and leg-length-normalized walking speed in ambulatory individuals with cerebral palsy.

Jul 2026
- Instrumented Walker Research for Smart Assistive Mobility Applications**
Tools: [MATLAB, Python, Arduino, OpenCap, Hardware Prototyping]
 - Designed and implemented a multimodal instrumented walker integrating IMUs, force-resistive sensors, and motion capture to conduct preliminary human-walker interaction research toward smart walker development
 - Developed a Kalman filtering-based sensor fusion pipeline for walker orientation estimation and constructed event-level directional symmetry metrics for sequencing-dependent coordination analysis

Feb 2026 
- Visual SLAM Deployment and Volumetric Mapping for Assistive Navigation Robot**
Tools: [ROS2, NVIDIA Isaac ROS, nvblox, Intel RealSense, Ubuntu, Shell Scripting]
 - Deployed and debugged an NVIDIA Isaac ROS visual SLAM and nvblox pipeline for real-time localization and 3D volumetric reconstruction using Intel RealSense depth cameras.

Jan 2026
- Post-Stroke Gait Analysis Research for Hip Exoskeleton Applications**
Tools: [MATLAB, Python, Shell Scripting, Opensim, VICON]
 - Python & MATLAB-based project that implements a shell scripted pipeline which processes post-stroke biomechanics data and executes correlation analyses on various biofeedback targets against hip joint work to fathom which should be used for hip exoskeleton applications
 - Implemented during undergraduate research at *Georgia Institute of Technology EPIC Lab*

Sep 2025 - Dec 2025 
- Gait Visual Biofeedback System for Hip Exoskeleton**
Tools: [Python, NVIDIA Jetson Orin Nano, UDP]
 - Python-based project that implements a real-time visual biofeedback module using IMU sensor data communicated from NVIDIA Jetson Orin Nano client to Laptop server via UDP protocol
 - Implemented during undergraduate research at *Georgia Institute of Technology EPIC Lab*

Sep 2025 - Nov 2025 
- Motion Detection Automated Parrot Shower System**
Tools: [ESP32, C++, CAD, 3D-Printing, Laser-cutter]
 - Hardware prototyping project that implements a motion-trigger sprinkler system using ESP32-CAM
 - Built a safe embedded hardware pipeline using NPN-driven relay switching, waterproof enclosures, and isolated low-voltage power to prevent hazards, all enclosed in a robust 3D-printed structure
 - Coursework for an Animal-Computer Interaction course (CS4803 at *Georgia Institute of Technology*)

Oct 2025 - Dec 2025 
- Embedded Systems & Electronics & Mechanics Skill Acquisition**
Tools: [Arduino, C/C++, Soldering, CAD, 3D-Printing, Laser-cutter]
 - Completed a sequence of embedded-systems labs covering microcontrollers, digital/analog sensing, ADC pipelines, PWM generation, motor control, communication protocols, and real-time interrupt-driven processing
 - Implemented circuits involving switches, photoresistors, potentiometers, microphones, diodes, capacitors, voltage regulators, transistors, and H-bridges
 - Designed and fabricated custom mechanical and electronic components, including CAD-modeled parts, 3D-printed components, and laser-cut structures, while building circuits via soldering, crimping, oscilloscope waveform inspection, and multimeter measurements
 - Coursework for a Prototyping Intelligent Devices course (CS3651 at *Georgia Institute of Technology*)

Oct 2025 - Dec 2025
- xv6 File System Extension Implementation**
Tools: [C, Linux, xv6, Git]
 - C-based project that extends the xv6 file system to support doubly-indirect blocks, raise the maximum file size from 1000 to 20000 blocks, and implemented bddr syscall to return the on-disk block information
 - Coursework for an Operating Systems course (CAS3101 at *Yonsei University*)

Jun 2025
- Principal Component Analysis(PCA) Image Reconstruction Implementation**
Tools: [Python, LaTeX]
 - Python-based project that implements conventional PCA for MNIST dataset image compression and reconstruction across varying numbers of principal components
 - Coursework for a Machine Learning course (CAS3120 at *Yonsei University*)

Jun 2025
- Image Segmentation via Kruskal's Algorithm**
Tools: [Python]

Jun 2025

- Python-based project that converts 2D images into weighted graphs—mapping each pixel to a vertex and connecting adjacent pixels with color-difference edge weights—and runs a modified Kruskal's algorithm that stops early to produce a spanning forest of a specified number of clusters, validated on small synthetic arrays and high-resolution real images.
- Coursework for a Data Structures course (CAS2103 at *Yonsei University*)
- **Accelerating RDBMS with CPU Index Filtering for Efficient GPU Query Execution** May 2025 - Jun 2025
Tools: [C++, CUDA, SQLite3] 
- C++-based project that implements a hybrid join engine: built a CPU-side hash-based filter to pre-select relevant rows and two CUDA kernels (counting and result-writing) that use atomicAdd and dynamic device memory allocation to safely process over 1 million record pairs without OOM(Out of Memory)
- Conducted a comprehensive performance study comparing three methods—pure CPU nested-loop join, brute-force GPU join, and hybrid CPU-GPU join—measuring microsecond-level runtimes and memory-transfer overheads
- Implemented during undergraduate capstone research at *Yonsei University HPCP Lab*
- **xv6 Thread Support Implementation** Jun 2025
Tools: [C, Linux, xv6, Git]
- C-based project that implements clone() and join() system calls in xv6 to create and synchronize lightweight threads sharing code, data, and file descriptors within a process, each with its own 4 KB-aligned stack and unique thread ID
- Developed thread_create() and thread_join() user-level library functions to allocate aligned stacks, invoke clone(), and manage thread lifecycle
- Added mutex_lock() and mutex_unlock() system calls to provide blocking mutex primitives for safe critical-section protection across threads
- Coursework for an Operating Systems course (CAS3101 at *Yonsei University*)
- **Gaussian Mixture Models & K-Means Clustering Implementation** May 2025
Tools: [Python, LaTeX]
- Python-based project implementing the EM algorithm for Gaussian Mixture Models—deriving and coding log-likelihood, posterior responsibilities (E-step), M-step update rules for means, variances, and mixing weights, plus analytical gradient computations—and writing K-Means clustering code to explore centroid initialization effects and apply clustering to image compression
- Coursework for a Machine Learning course (CAS3120 at *Yonsei University*)
- **Image Compression via Quadtree Data Structure** May 2025
Tools: [Python]
- Python-based project that implements recursive quadtree algorithms for threshold-based image compression and validates them on 256×256 real images
- Coursework for a Data Structures course (CAS2103 at *Yonsei University*)
- **xv6 Virtual Memory & Demand Paging & Copy-on-Write Implementation** May 2025
Tools: [C, Linux, xv6, Git]
- C-based project that implements mmap() and munmap() system calls to xv6—handling 4 KB-aligned, file-backed mappings with proper permission flags, dirty-page write-back, and cleanup on fork, close, and exit
- Implemented demand paging by marking newly allocated regions as non-present and extending the page-fault handler to allocate and map physical pages on first access, reducing startup overhead and memory footprint
- Extended fork() to support copy-on-write semantics by marking shared pages read-only, implementing a page-fault handler to clone pages, and maintaining per-frame reference counts for safe memory reclamation
- Coursework for an Operating Systems course (CAS3101 at *Yonsei University*)
- **Support Vector Machine & Neural Network Implementation** Apr 2025
Tools: [Python, LaTeX]
- Python-based project that derives and implements hard- and soft-margin SVM formulations—writing out all constraints, solving for optimal parameters, and computing analytical gradients for Bernoulli data classification
- Implemented forward- and backward-propagation routines of a neural network and ran extensive experiments to analyze the effects of hyperparameter and model architecture adjustments on its performance
- Coursework for a Machine Learning course (CAS3120 at *Yonsei University*)
- **xv6 CPU EEVDF Scheduler Implementation** Apr 2025
Tools: [C, Linux, xv6, Git]
- C-based project that develops an EEVDF (Earliest Eligible Virtual Deadline First) scheduler in xv6 by computing virtual time, eligible time, virtual deadlines, and lag (with integer scaling) along with test programs to achieve fair time-share scheduling
- Coursework for an Operating Systems course (CAS3101 at *Yonsei University*)

- **xv6 CPU Priority Scheduler Implementation**

Tools: [C, Linux, xv6, Git]

 - C-based project that implements a weight-and-PID-based priority scheduler along with test programs in xv6 using `sched_setattr` to assign weights among RUNNABLE processes
 - Coursework for an Operating Systems course (CAS3101 at *Yonsei University*)

Apr 2025
- **Content-Aware Image Resizing Algorithm Implementation**

Tools: [Python, SciPy]

 - Python-based project that converts images into 2D arrays then into 2D doubly linked lists and implements recursive algorithms for seam carving to achieve context-aware image resizing
 - Coursework for a Data Structures course (CAS2103 at *Yonsei University*)

Apr 2025
- **Machine Learning Algorithm Implementation**

Tools: [Python, Scikit-learn, LaTeX]

 - Python-based project that implements key machine learning algorithms—linear regression, logistic regression, gradient descent, and stochastic gradient descent—while documenting theoretical derivations and empirical analyses compiled in LaTeX
 - Coursework for a Machine Learning course (CAS3120 at *Yonsei University*)

Apr 2025
- **Extended System Calls in xv6 Operating System**

Tools: [C, Linux, xv6, Git]

 - C-based project that implements four new xv6 system calls — `sched_setattr`, `sched_getattr`, `yield`, and `ps` — to enable user-level control and introspection of process scheduling attributes (requested CPU ticks and weight)
 - Coursework for an Operating Systems course (CAS3101 at *Yonsei University*)

Mar 2025
- **Machine Learning Mathematical Analysis Implementation**

Tools: [Python, LaTeX]

 - Python-based project that implements correlation-based and rank-based independence check methods
 - LaTeX-compiled project that develops mathematical proofs for core principles in probability, statistics, linear algebra, and calculus
 - Coursework for a Machine Learning course (CAS3120 at *Yonsei University*)

Mar 2025
- **Audio Classification AI & Android Mobile Application**

Tools: [Python, TensorFlow Lite, YAMNet, Kotlin]

 - Python-based project that develops custom gunshot and explosion audio classification AI based on YAMNet model using TensorFlow Lite framework to integrate it into a mobile application
 - Kotlin-based project that enables the usage of the custom audio classification AI on an android app
 - Implemented during 2024 Republic of Korea Army Startup Competition as role of CTO

May 2024 - Aug 2024

[🌐]
- **Improved Five-Stage Pipelined MIPS CPU Implementation**

Tools: [C++, Linux]

 - C++-based project that implements two hardware-level improvements to a vanilla five-stage pipelined MIPS CPU, utilizing data forwarding and hazard detection techniques to facilitate execution of data-dependent instructions
 - Coursework for an Architecture of Computers course (CSI3102 at *Yonsei University*)

Jun 2023
- **Vanilla Five-Stage Pipelined MIPS CPU Implementation**

Tools: [C++, Linux]

 - C++-based project that implements a five-stage pipelined MIPS CPU, utilizing a datapath and control units from a single-cycle MIPS CPU with the inclusion of bubbles to handle stalls and maintain functional correctness
 - Coursework for an Architecture of Computers course (CSI3102 at *Yonsei University*)

Jun 2023
- **Single-Cycle MIPS CPU Implementation**

Tools: [C++, Linux]

 - C++-based project that implements a functional single-cycle MIPS CPU with various hardware components, including a single-cycle datapath and control units, supporting ten MIPS instructions
 - Coursework for an Architecture of Computers course (CSI3102 at *Yonsei University*)

May 2023
- **MIPS Assembly Language Implementation**

Tools: [MIPS, SPIM]

 - MIPS-based project that implements Fibonacci with error handling, character index search, recursive GCD/LCM and in-place quickSort.
 - Coursework for an Architecture of Computers course (CSI3102 at *Yonsei University*)

Apr 2023
- **Continual Learning Image Classification Project**

Tools: [Python, PyTorch, Avalanche]

 - Python-based deep learning model using CNN, PyTorch and Avalanche library to incrementally learn custom CIFAR-10 data across 5 consecutive tasks of recognizing 2 classes of images, resulting in a solution for continual learning of video data

May 2023

[🌐]

◦ Coursework for an Artificial Intelligence course (CSI4108 at *Yonsei University*)

• **Neuron Action Potential Propagation Simulation**

Mar 2023 - Apr 2023

Tools: [MATLAB]



- MATLAB-based codes that implement the leaky integrate-and-fire model (LIF model), exponential leaky integrate-and-fire model (ELIF model), adaptive exponential leaky integrate-and-fire model (AELIF model), and Hodgkin-and-Huxley model

- Implemented during undergraduate research at *Yonsei University cNeuro Lab*

• **Low-Cost Airlines Passenger Data Analysis**

Feb 2023

Tools: [Python]



- Python-based web-scraping code for mining key words that may suggest new business tactics
- Python-based code that preprocesses then analyzes open-source passengers' data
- Implemented for a Python Data Analysis Competition held by *Yonsei University*

• **Moving-Target Acquisition Experiment Statistical Analysis**

Dec 2022

Tools: [JASP]

- Statistical analysis of an HCI experiment using JASP such as null hypothesis testing (repeated-measures ANOVA), assumption checks (sphericity tests, homogeneity tests), and post-hoc analysis (Bonferroni correction)
- Coursework for a Human-Computer Interface course (CSI4107 at *Yonsei University*)

• **Gaze-and-Cursor-Driven Secondary Display Position Estimation**

Nov 2022

Tools: [Python, Scipy]

- Python-based project that uses gaze tracking and cursor movement on a dual-display setup to estimate the position of a secondary display
- Coursework for a Human-Computer Interface course (CSI4107 at *Yonsei University*)

• **Moving-Target Acquisition Model Fitting**

Nov 2022

Tools: [Python, Scikit-learn]

- Python-based code that fits the experimental dataset to a model for moving-target acquisition experiment using the least squares method and returns a root mean squared error rate using scikit-learn library
- Python-based code that estimates the model parameter values by fitting the Fitts' law equation to the experimental dataset using scikit-learn library
- Python-based code that estimates the model parameter values by fitting the drift-diffusion model to the experimental dataset and also return the log-likelihood value of the estimation (maximum likelihood estimation) using scikit-learn
- Coursework for a Human-Computer Interface course (CSI4107 at *Yonsei University*)

• **Moving-Target Acquisition Experiment Implementation**

Oct 2022

Tools: [Python, Scikit-learn]



- Processing project implemented with Java that implements a moving target acquisition experiment
- Coursework for a Human-Computer Interface course (CSI4107 at *Yonsei University*)

• **KakaoTalk Messenger Conversation Analyzer**

Dec 2021

Tools: [Python]



- Python-based project that analyzes the participants' activity in KakaoTalk messenger chat
- Final project for a Python programming course (SED2005 *Yonsei University*)

SKILLS

- **Programming:** Python, MATLAB, C, C++, Java, MIPS, Swift, Kotlin, SQL
- **Hardware:** NVIDIA Jetson, CAD, Arduino, Soldering, 3D-Printing, Laser-cutter, COSMED
- **Technical Software:** OpenSim, Vicon, OpenCap, LaTeX, ROS 2, Processing, JASP, Git, Linux, CUDA, Docker
- **Data Science & Machine Learning:** TensorFlow, PyTorch, Scikit-learn, SciPy, Avalanche

PROFESSIONAL MEMBERSHIPS

- **Korean-American Scientists and Engineers Association (KSEA)**
- **Korean Society of Human Brain Mapping (KHBM)**

Sep 2025 - Present

Jan 2023 - Present

CERTIFICATIONS

- **Republic of Korea's Ministry of Science and ICT:** Craftsman Information Processing Apr 2025
- **Commissioner of Seoul, Metropolitan Police Agency:** Driver's License – 1st Class (Ordinary) Feb 2021

LANGUAGES

- **English(Native):** TOEFL(118/120), TOEIC(990/990), NEW TEPS(556/600)
- **Korean(Mother-Tongue)**
- **French(Limited Working Proficiency):** DELF B2

REFERENCES

1. **M.D., Ph.D. Dongwook Rha** [🌐]
Professor, Department of Rehabilitation Medicine / College of Medicine / Yonsei University
Email: MEDICUS@yuhs.ac
Relationship: Master's Advisor
2. **Ph.D. Jongmin Lee** [🌐]
Assistant Professor, Department of Artificial Intelligence / College of Computing / Yonsei University
Email: jongminlee@yonsei.ac.kr
Relationship: Master's Co-Advisor
3. **Ph.D. Aaron Young** [🌐]
Associate Professor, School of Mechanical Engineering / College of Engineering / Georgia Institute of Technology
Email: aaron.young@me.gatech.edu
Relationship: Undergraduate Research Assistantship Advisor